

SQL Case Study – 2

Problem Statement:

You are working as a database administrator for an organization that maintains employee records, departmental information, job roles, and office locations in a relational database system. The management team wants to analyze employee-related data to better understand salary distribution, departmental performance, hiring trends, and workforce allocation across different locations.

To support these business requirements, multiple tables such as Employee, Department, Job, and Location have been created. Using SQL queries, you are required to retrieve, manipulate, and analyze data from these tables. The case study focuses on applying various SQL concepts including simple queries, filtering conditions, sorting, grouping, aggregate functions, joins, conditional statements, and subqueries.

The objective of this case study is to gain practical understanding of how SQL can be used to manage organizational data efficiently, generate meaningful reports, and support data-driven decision-making processes within an enterprise environment.

Data Sets:

Create the following tables:

LOCATION	
Location_ID(PK)	City
122	New York
123	Dallas
124	Chicago
167	Bostan

DEPARTMENT		
Department_id(PK)	Name	Location_Id(FK)
10	Accounting	122
20	Sales	124
30	Research	123
40	Operations	167

JOB	
Job_ID(PK)	Designation
667	Clerk
668	Staff
669	Analyst
670	Sales Person
671	Manager
672	President

EMPLOYEE								
Employee_Id	Last_Name	First_Name	Middle_Name	Job_Id(FK)	Hire Date	Salary	Comm	Department_Id(FK)
7369	Smith	John	Q	667	17-Dec-84	800	Null	20
7499	Allen	Kevin	J	670	20-Feb-85	1600	300	30
755	Doyle	Jean	K	671	04-Apr-85	2850	Null	30
756	Dennis	Lynn	S	671	15-May-85	2750	Null	30
757	Baker	Leslie	D	671	10-Jun-85	2200	Null	40
7521	Wark	Cynthia	D	670	22-Feb-85	1250	50	30

Task to be Performed:

Simple Queries:

1. List all the employee details.

```
--1. List all the employee details
Select * from [dbo].[EMPLOYEE];
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
2	7499	ALLEN	KEVIN	J	670	7698	1984-02-20	1600	300	30
3	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
4	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
5	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40
6	7521	WARK	CYNTHIA	D	670	7698	1985-02-22	1250	500	30

2. List all the department details.

```
--2. List all the department details
Select * from [dbo].[DEPARTMENT];
```

Output

	Department_Id	Name	Location_Id
1	10	Accounting	122
2	20	Sales	124
3	30	Research	123
4	40	Operations	167

3. List all job details.

```
--3. List of all job details.  
Select * from [dbo].[JOB];
```

Output

	Job_ID	Designation
1	667	CLERK
2	668	STAFF
3	669	ANALYST
4	670	SALES_PERSON
5	671	MANAGER
6	672	PRESIDENT

4. List all the locations.

```
--4. List of all the location.  
Select * from [dbo].[LOCATION];
```

Output

	Location_ID	City
1	122	New York
2	123	Dallas
3	124	Chicago
4	167	Boston

5. List out the First Name, Last Name, Salary, Commission for all Employees.

```
--5. List out the first name, last name, salary, commission for all employee  
Select [First_Name], [Last_Name], [Salary], [COMM]  
From [dbo].[EMPLOYEE];
```

Output

	First_Name	Last_Name	Salary	COMM
1	JOHN	SMITH	800	NULL
2	KEVIN	ALLEN	1600	300
3	JEAN	DOYLE	2850	NULL
4	LYNN	DENNIS	2750	NULL
5	LESLIE	BAKER	2200	NULL
6	CYNTHIA	WARK	1250	500

6. List out the Employee ID, Last Name, Department ID for all employees and alias - Employee ID as "ID of the Employee", Last Name as "Name of the Employee", Department ID as "Dep_id".

```
--6. List out the employee ID, last name, department Id for all the employees and alias  
Select [Employee_Id] AS [ID of the Employee], [Last_Name] AS [Name of the Employee], [Department_Id] AS Dep_Id  
From [dbo].[EMPLOYEE];
```

Output

	ID of the Employee	Name of the Employee	Dep_Id
1	7369	SMITH	20
2	7499	ALLEN	30
3	7505	DOYLE	30
4	7506	DENNIS	30
5	7507	BAKER	40
6	7521	WARK	30

- List out the annual salary of the employees with their names only.

--7. List out the annual salary of the employee with their names only.

```
Select [First_Name] + ' ' + [Last_Name] AS [Name], [Salary] * 12 AS Annual_Salary
From [dbo].[EMPLOYEE];
```

Output

	Name	Annual_Salary
1	JOHN SMITH	9600
2	KEVIN ALLEN	19200
3	JEAN DOYLE	34200
4	LYNN DENNIS	33000
5	LESLIE BAKER	26400
6	CYNTHIA WARK	15000

WHERE Condition:

- List the details about "Smith".

--1. List the details about Smith

```
Select * from [dbo].[EMPLOYEE]
Where [Last_Name] = 'Smith';
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20

- List out the employees who are working in department 20.

--2. List out the employee who are working in department 20.

```
Select * from [dbo].[EMPLOYEE]
Where [Department_Id] = 20;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20

- List out the employees who are earning salary between 2000 and 3000.

--3. List out all the employees who are earning between 2000 and 3000

```
Select * from [dbo].[EMPLOYEE]
Where [Salary] BETWEEN 2000 AND 3000;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
2	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
3	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

- List out the employees who are working in department 10 or 20.

--4. list out all the employees who are working in department 10 or 20.

```
Select * from [dbo].[EMPLOYEE]
Where [Department_Id] IN (10,20);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20

5. Find out the employees who are not working in department 10 or 30.

--5. Find out the employees who are not working in department 10 or 30.

```
Select * from [dbo].[EMPLOYEE]
Where [Department_Id] NOT IN (10,30);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
2	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

6. List out the employees whose name starts with 'L'.

--6. List out the employees whose name start with L.

```
Select * from [dbo].[EMPLOYEE]
Where [First_Name] LIKE 'L%';
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
2	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

7. List out the employees whose name starts with 'L' and ends with 'E'.

--7. List out the employees whose name starts with L and ends with E.

```
Select * from [dbo].[EMPLOYEE]
Where [First_Name] LIKE 'L%e';
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

8. List out the employees whose name length is 4 and start with 'J'.

--8. List out the employees whose name length is 4 and start with J.

```
Select * from [dbo].[EMPLOYEE]
Where LEN([First_Name]) = 4 AND [First_Name] LIKE 'J%';
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
2	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30

9. List out the employees whose name length is 4 and start with 'J'. List out the employees who are working in department 30 and draw the salaries more than 2500.

--9. List out the employees who are working in department 30 and draw salary more than 2500.

```
Select * from [dbo].[EMPLOYEE]
Where [Department_Id] = 30 AND [Salary] > 2500;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
2	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30

10. List out the employees who are not receiving commission.

```
--10. List out the employee who are not receiving commission.  
Select * from [dbo].[EMPLOYEE]  
Where [COMM] IS NULL;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
2	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
3	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
4	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

ORDER BY Clause:

1. List out the Employee ID and Last Name in ascending order based on the Employee ID.

```
--1. List out employee ID and last name in ascending order based on employee id.  
Select [Employee_Id], [Last_Name]  
From [dbo].[EMPLOYEE]  
ORDER BY [Employee_Id];
```

Output

	Employee_Id	Last_Name
1	7369	SMITH
2	7499	ALLEN
3	7505	DOYLE
4	7506	DENNIS
5	7507	BAKER
6	7521	WARK

2. List out the Employee ID and Name in descending order based on salary.

```
--2. List out the employee ID and Name in descending order based on salary.  
Select [Employee_Id], [Last_Name]  
From [dbo].[EMPLOYEE]  
ORDER BY [Salary] DESC;
```

Output

	Employee_Id	Last_Name
1	7505	DOYLE
2	7506	DENNIS
3	7507	BAKER
4	7499	ALLEN
5	7521	WARK
6	7369	SMITH

- List out the employee details according to their Last Name in ascending-order.

```
--3. List out the employee details according to their last name in ascending order.
Select * From [dbo].[EMPLOYEE]
ORDER BY [Last_Name];
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7499	ALLEN	KEVIN	J	670	7698	1984-02-20	1600	300	30
2	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40
3	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
4	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
5	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
6	7521	WARK	CYNTHIA	D	670	7698	1985-02-22	1250	500	30

- List out the employee details according to their Last Name in ascending order and then Department ID in descending order.

```
--4. List out the employee details according to their last name in ascending order and then department id in descending order.
Select * From [dbo].[EMPLOYEE]
ORDER BY [Last_Name] ASC, [Department_Id] DESC;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7499	ALLEN	KEVIN	J	670	7698	1984-02-20	1600	300	30
2	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40
3	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30
4	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
5	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20
6	7521	WARK	CYNTHIA	D	670	7698	1985-02-22	1250	500	30

GROUP BY and HAVING Clause:

- List out the department wise maximum salary, minimum salary and average salary of the employees.

```
--1. List out the department wise maximum salary, minimum salary and average salary of the employee.
Select [Department_Id], Max([Salary]) AS Max_Salary, Min([Salary]) AS Min_salary, AVG([Salary]) AS Average_Salary
From [dbo].[EMPLOYEE]
GROUP BY [Department_Id];
```

Output

	Department_Id	Max_Salary	Min_salary	Average_Salary
1	20	800	800	800
2	30	2850	1250	2112
3	40	2200	2200	2200

- List out the job wise maximum salary, minimum salary and average salary of the employees.

```
--2. List out the job wise maximum salary, minimum salary and average salary of the employee.
Select [Job_Id], Max([Salary]) AS Max_Salary, Min([Salary]) AS Min_salary, AVG([Salary]) AS Average_Salary
From [dbo].[EMPLOYEE]
GROUP BY [Job_Id];
```

Output

	Job_Id	Max_Salary	Min_salary	Average_Salary
1	667	800	800	800
2	670	1600	1250	1425
3	671	2850	2200	2600

3. List out the number of employees who joined each month in ascending order.

```
--3. List out the number of employee who joined each month in ascending order.  
Select Month([Hire_Date]) AS MONTH, COUNT(*) AS EmployeeCount  
FROM [dbo].[EMPLOYEE]  
GROUP BY [Hire_Date]  
ORDER BY MONTH;
```

Output

	MONTH	EmployeeCount
1	2	1
2	2	1
3	4	1
4	5	1
5	6	1
6	12	1

4. List out the number of employees for each month and year in ascending order based on the year and month.

```
--4. List out the number of employee for each month and year in the ascending based on year and month  
Select MONTH([Hire_Date]) AS Month, YEAR([Hire_Date]) AS Year, Count(*) AS EmployeeCount  
From [dbo].[EMPLOYEE]  
Group by YEAR([Hire_Date]), MONTH([Hire_Date])  
Order by Year, Month;
```

Output

	Month	Year	EmployeeCount
1	2	1984	1
2	12	1984	1
3	2	1985	1
4	4	1985	1
5	5	1985	1
6	6	1985	1

5. List out the Department ID having at least four employees.

```
--5. List out the Department Id having atleast four employee.  
Select [Department_Id]  
From [dbo].[EMPLOYEE]  
GROUP BY [Department_Id]  
Having Count(*)>=4;
```

Output

	Department_Id
1	30

6. How many employees joined in February month.

```
--6. How many employees joining in February month  
Select Count(*) AS Feb_EMP  
From [dbo].[EMPLOYEE]  
Where MONTH([Hire_Date]) = 2;
```

Output

	Feb_EMP
1	2

7. How many employees joined in May or June month.

```
--7. How many employee joined in May or June Month
Select Count(*) AS Feb_Jun_Emp
From [dbo].[EMPLOYEE]
Where MONTH([Hire_Date]) In (5,6);
```

Output

	Feb_Jun_Emp
1	2

8. How many employees joined in 1985?

```
--8. How many employee joined in 1985
Select Count(*) AS Joined_1985
From [dbo].[EMPLOYEE]
where Year([Hire_Date]) = 1985;
```

Output

	Joined_1985
1	4

9. How many employees joined each month in 1985?

```
--9. How many employee joined each month in 1985
Select Count(*) AS Emp_No, MONTH([Hire_Date]) AS Month
From [dbo].[EMPLOYEE]
where YEAR([Hire_Date]) = 1985
GROUP BY MONTH([Hire_Date]);
```

Output

	Emp_No	Month
1	1	2
2	1	4
3	1	5
4	1	6

10. How many employees were joined in April 1985?

```
--10. How many employee joined in April 1985.
Select Count(*) AS Emp_joined_Apr_1985
From [dbo].[EMPLOYEE]
Where YEAR([Hire_Date]) = 1985 AND MONTH([Hire_Date]) = 4;
```

Output

	Emp_joined_Apr_1985
1	1

JOINS:

1. List out employees with their department names.

--1. List out employees with their department name

```
Select E.[First_Name] AS Emp_Name, D.[Name] AS Dept_Name  
From [dbo].[EMPLOYEE] E  
Join [dbo].[DEPARTMENT] D on E.[Department_Id] = D.[Department_Id];
```

Output

	Emp_Name	Dept_Name
1	JOHN	Sales
2	KEVIN	Research
3	JEAN	Research
4	LYNN	Research
5	LESLIE	Operations
6	CYNTHIA	Research

2. Display employees with their designations.

--2. Display Employees with their designation

```
Select E.[First_Name] AS Emp_Name, J.[Designation] AS Designation  
From [dbo].[EMPLOYEE] E  
Join [dbo].[JOB] J on E.[Job_Id] = J.[Job_ID];
```

Output

	Emp_Name	Designation
1	JOHN	CLERK
2	KEVIN	SALES_PERSON
3	JEAN	MANAGER
4	LYNN	MANAGER
5	LESLIE	MANAGER
6	CYNTHIA	SALES_PERSON

3. Display the employees with their department names and city.

--3. Display Employee with their department name and city

```
Select E.[First_Name], D.[Name], L.[City]  
From [dbo].[EMPLOYEE] E  
Join [dbo].[DEPARTMENT] D on E.Department_Id = D.Department_Id  
Join [dbo].[LOCATION] L on D.Location_Id = L.Location_ID;
```

Output

	First_Name	Name	City
1	JOHN	Sales	Chicago
2	KEVIN	Research	Dallas
3	JEAN	Research	Dallas
4	LYNN	Research	Dallas
5	LESLIE	Operations	Boston
6	CYNTHIA	Research	Dallas

4. How many employees are working in different departments? Display with department names.

```
--4. How many employees are working in different departments? Display with department name.  
Select D.[Name], Count(E.[First_Name]) AS No_of_Emp  
From [dbo].[EMPLOYEE] E  
Join [dbo].[DEPARTMENT] D on E.Department_Id = D.Department_Id  
Group BY D.Name;
```

Output

	Name	No_of_Emp
1	Operations	1
2	Research	4
3	Sales	1

5. How many employees are working in the sales department?

```
--5. how many employees working in sales  
Select COUNT(*) AS Sales_Emp  
From [dbo].[EMPLOYEE] E  
Join [dbo].[DEPARTMENT] D on E.Department_Id = D.Department_Id  
Where D.[Name] = 'Sales';
```

Output

	Sales_Emp
1	1

6. Which is the department having greater than or equal to 3 employees and display the department names in ascending order.

```
--6. Which is the department having greater than or equal to 3 employees and display the department name in ascending order.  
Select E.[First_Name] AS Emp_Name, D.[Name] AS Dept_Name  
From EMPLOYEE E  
Join [dbo].[DEPARTMENT] D on E.Department_Id = D.Department_Id  
Order by D.[Name];
```

Output

	Emp_Name	Dept_Name
1	LESLIE	Operations
2	KEVIN	Research
3	JEAN	Research
4	LYNN	Research
5	CYNTHIA	Research
6	JOHN	Sales

7. How many employees are working in 'Dallas'?

```
--7. How many employees working from Dallas  
Select COUNT(*) AS Total_Emp  
From EMPLOYEE E  
Join [dbo].[DEPARTMENT] D on E.[Department_Id] = D.Department_Id  
Join [dbo].[LOCATION] L on D.Location_Id = L.Location_ID  
Where [City] = 'Dallas';
```

Output

	Total_Emp
1	4

- Display all employees in sales or operation departments.

```
--8. Display all the employees in sales or operation department
Select E.[First_Name] AS emp_Name, D.[Name] AS Dept_Name
From EMPLOYEE E
Join [dbo].[DEPARTMENT] D on E.Department_Id = D.Department_Id
Where D.[Name] in ('Sales', 'Operations');
```

Output

	emp_Name	Dept_Name
1	JOHN	Sales
2	LESLIE	Operations

CONDITIONAL STATEMENT

- Display the employee details with salary grades. Use conditional statement to create a grade column.

```
--1. Display the employee details with salary grade. Use conditional statement to create a grade column.
Select *,
CASE
    When [Salary] < 1000 THEN 'Grade D'
    When [Salary] BETWEEN 1000 AND 1999 THEN 'Grade C'
    When [Salary] Between 2000 AND 2999 THEN 'Grade B'
    When [Salary] >= 3000 THEN 'Grade A'
    ELSE 'No Grade'
End AS Grade
From [dbo].[EMPLOYEE];
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID	Grade
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20	Grade D
2	7499	ALLEN	KEVIN	J	670	7698	1984-02-20	1600	300	30	Grade C
3	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30	Grade B
4	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30	Grade B
5	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40	Grade B
6	7521	WARK	CYNTHIA	D	670	7698	1985-02-22	1250	500	30	Grade C

- List out the number of employees grade wise. Use conditional statement to create a grade column

```
--2. List out the number of employees grade wise. Use conditional statement to create a grade column.
Select
CASE
    When [Salary] < 1000 THEN 'Grade D'
    When [Salary] BETWEEN 1000 AND 1999 THEN 'Grade C'
    When [Salary] Between 2000 AND 2999 THEN 'Grade B'
    When [Salary] >= 3000 THEN 'Grade A'
    ELSE 'No Grade'
End AS Grade,
Count(*) AS Emp_Count
From [dbo].[EMPLOYEE]
Group By
CASE
    When [Salary] < 1000 THEN 'Grade D'
    When [Salary] BETWEEN 1000 AND 1999 THEN 'Grade C'
    When [Salary] Between 2000 AND 2999 THEN 'Grade B'
    When [Salary] >= 3000 THEN 'Grade A'
    ELSE 'No Grade'
End;
```

Output

	Grade	Emp_Count
1	Grade B	3
2	Grade C	2
3	Grade D	1

- Display the employee salary grades and the number of employees between 2000 to 5000 range of salary.

--3. Display the employee salary grade and tge number of employees between 200 to 5000 range of salary.

```
SELECT
CASE
    WHEN Salary BETWEEN 2000 AND 2999 THEN 'Grade B'
    WHEN Salary BETWEEN 3000 AND 5000 THEN 'Grade A'
    ELSE 'Other'
END AS Grade,
COUNT(*) AS EmployeeCount
FROM Employee
WHERE Salary BETWEEN 2000 AND 5000
GROUP BY
CASE
    WHEN Salary BETWEEN 2000 AND 2999 THEN 'Grade B'
    WHEN Salary BETWEEN 3000 AND 5000 THEN 'Grade A'
    ELSE 'Other'
END;
```

Output

	Grade	EmployeeCount
1	Grade B	3

SUBQUERIES:

- Display the employees list who got the maximum salary.

--1. Display the employees list who got the maximum salary.

```
Select * From [dbo].[EMPLOYEE]
Where [Salary] = (Select Max([Salary]) From [dbo].[EMPLOYEE]);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30

- Display the employees who are working in the sales department.

--2. Display the employees who are working in th sales department

```
Select * From [dbo].[EMPLOYEE]
WHERE [Department_Id] = (Select [Department_Id] From [dbo].[DEPARTMENT] Where [Name] = 'Sales');
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20

- Display the employees who are working as 'Clerk'.

--3. Display the Employee who are working as Clerk

```
Select * From [dbo].[EMPLOYEE]
Where Job_Id = (Select [Job_ID] From [dbo].[JOB] Where [Designation] = 'Clerk');
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7369	SMITH	JOHN	Q	667	7902	1984-12-17	800	NULL	20

4. Display the list of employees who are living in 'Boston'.

```
--4. Display the list of Employees who lives in Boston.
Select * From [dbo].[EMPLOYEE]
Where [Department_Id] = (Select [Department_Id] From [dbo].[DEPARTMENT]
Where Location_Id = (Select [Location_ID] FROM [dbo].[LOCATION] Where [City] = 'Boston')
);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7507	BAKER	LESLIE	D	671	7839	1985-06-10	2200	NULL	40

5. Find out the number of employees working in the sales department.

```
--5. Find out the number of employees working in sales department.
Select Count(*) From [dbo].[EMPLOYEE]
Where [Department_Id] = (Select [Department_Id] From [dbo].[DEPARTMENT] Where [Name] = 'Sales' );
```

Output

	(No column name)
1	1

6. Update the salaries of employees who are working as clerks on the basis of 10%.

```
--6. Update the salaries of the employees who are working as clerks on the basis of 10%
UPDATE EMPLOYEE
Set Salary = Salary * 1.1
Where Job_Id = (Select Job_Id From JOB Where Designation = 'Clerk');
```

Output

(1 row affected)

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7. Display the second highest salary drawing employee details.

```
--7. Display the second highest salary drawing employee details.
Select TOP 1*
From EMPLOYEE
Where Salary < (Select Max(Salary) From EMPLOYEE)
Order by Salary DESC;
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30

8. List out the employees who earn more than every employee in department 30.

```
--8. List out the employees who earn more than every employee in department 30
Select * from EMPLOYEE
Where Salary > ALL (Select Salary From [dbo].[EMPLOYEE] Where Department_Id = 30);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
--	-------------	-----------	------------	-------------	--------	------------	-----------	--------	------	---------------

9. Find out which department has no employees.

--9.Find Out which department has no employees

```
Select [Department_Id],[Name] from [dbo].[DEPARTMENT]
Where [Department_Id] NOT IN (Select Distinct [Department_Id] From EMPLOYEE);
```

Output

	Department_Id	Name
1	10	Accounting

10. Find out the employees who earn greater than the average salary for their department.

--10.Find out the employee who earns greater than the average salary from ther department

```
SELECT *
FROM EMPLOYEE E
WHERE Salary > (SELECT AVG(Salary) FROM EMPLOYEE WHERE Department_Id = E.Department_Id);
```

Output

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMM	DEPARTMENT_ID
1	7505	DOYLE	JEAN	K	671	7839	1985-04-04	2850	NULL	30
2	7506	DENNIS	LYNN	S	671	7839	1985-05-15	2750	NULL	30

Conclusion:

This SQL case study provided comprehensive hands-on experience in managing and analyzing organizational data using SQL. By working with employee, department, job, and location tables, the study demonstrated how relational databases help store, retrieve, and process business information efficiently.

Throughout the case study, several important SQL concepts were implemented, including SELECT statements, filtering using WHERE conditions, sorting with ORDER BY, grouping and aggregation using GROUP BY and HAVING clauses, joins for combining multiple tables, conditional statements for categorization, and subqueries for advanced data retrieval. These operations enabled detailed analysis of employee salaries, departmental workforce distribution, hiring patterns, job roles, and location-based employee information.

The case study highlighted the importance of SQL in:

Managing employee and departmental records effectively

Generating business reports and summaries

Performing salary and performance analysis

Understanding workforce distribution across departments and locations

Supporting organizational decision-making through structured data analysis

Overall, this case study strengthened the understanding of relational database operations and demonstrated how SQL serves as a powerful tool for handling real-world business data and solving practical organizational problems efficiently.